

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective removes most of the objective available to it and keeps the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the structure it

is asked to learn, and further from it than from almost every relabelling of the nodes; two controls sharing its step size, the second of them also its cumulative kernel deformation, do neither. Descending that functional does not produce representational learning, and its optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows.

12 Coupling must be strong enough to sustain coherent collective activity; the
13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the

sweep of Section 4, so an exponent fitted to a recovery trace measures the ramp at least as much as the mechanism—and that holds for any analysis reading a critical exponent off a driven transition, not only for ours. The third is a negative result about plasticity: a coupling update descending a squared-drift objective removes most of the objective available to it and preserves coherence, while moving *away* from the environmental structure it is expected to learn—below almost every node relabelling, and beyond what either of two controls does: a random update of the same step size, and that same gradient step delivered to relabelled nodes, which reproduces its cumulative kernel deformation. Descending a squared-drift functional does not produce representational learning, and the direction of that failure follows from a conservation law rather than from the particular run. The fourth is a machine-checked composition (Section 5) in which finite information capacity, Landauer’s bound, a predictive dissipation bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem yield one conditional theorem, with the eight hypotheses that composition consumes isolated and named; one witness assembled from distinct toy components satisfies all eight at once, so their conjunction is not empty.

A prose table of assumptions would list the same eight items, so the fourth contribution needs a word about method. What the Lean development adds is that it makes the list non-negotiable. The characteristic failure in an argument of this shape is not a false step but a silent change of subject: a theorem about a stationary density read as a claim about a trajectory,

58 an inequality about a register read as a claim about learning, agreement of
59 phases read as compatibility of representations. Type-checking forbids each
60 of those substitutions, and an assumption needed to make a step go through
61 appears as a hypothesis in the term rather than as a word in a sentence. The
62 last of them is not merely forbidden but refuted by a constructed object: the
63 development carries a cover at a single common phase whose local densities
64 disagree where its patches meet, satisfying every other requirement of the
65 gluing theorem and failing compatibility alone (Section 5.2). That object
66 is what a prose table of assumptions cannot carry. Formalization does not
67 make the assumptions true; it makes them impossible to lose.

68 The physical model and the formal composition belong in one article be-
69 cause they are two halves of one commitment. The coupling-gated prediction
70 is worth testing under this framework, rather than as one more synchroniza-
71 tion model, only because the coherent state it produces is claimed to do
72 further work: to support a globally consistent representation and a reflexive
73 fixed point that we identify with phenomenal unity and a minimal self.

74 The distinction that does that work is between phases agreeing and de-
75 scriptions fitting together, and it is concrete. Take two cortical regions whose
76 receptive territories overlap, each carrying a probability distribution over
77 what occupies the shared part. Their oscillators can be locked to the same
78 phase at every instant while one region's distribution on the overlap places an
79 object left of fixation and the other places it right. The phases agree; the de-
80 scriptions do not, and no single distribution over the shared territory restricts

81 to both. A theory that identifies unity with synchrony cannot mark that dif-
82 ference, and so cannot separate a brain whose regions oscillate together from
83 one whose regions are about the same world. The sheaf condition marks
84 exactly it: local states glue to one global section when their restrictions to
85 shared regions coincide, and otherwise not at all. Coherence enters as the
86 physical condition under which compatibility can be maintained, not as a
87 substitute for it, and the self-prediction map then asks the further ques-
88 tion of whether the glued state contains an accurate model of itself. That
89 identification is a commitment, not a theorem, and Section 6 states it, ar-
90 gues for it, and says what would count against it. Its separation from the
91 physical tests is what makes both assessable: the cortical coupling hypothe-
92 sis and the recovery prediction can fail while the mathematical composition
93 stands, and the composition can stand while the identification is wrong. The
94 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
95 Banach, 1922) therefore appear here for their biological composition, not for
96 any new fundamental physical law.

2 A coupling-gated model of cortical coherence

2.1 Dynamics and stationary solutions

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly

113 increasing with K .

114 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

115 This is a result about the stationary solution branch. Of the coherent branch
116 we prove existence, uniqueness, continuous emergence at threshold and strict
117 growth in K , but not that it is dynamically selected. The dynamical mean-
118 field limit—propagation of chaos—is the missing link between the finite sys-
119 tem and the stationary description (Sznitman, 1991), and no theorem in this
120 development closes it. Exchange symmetry alone does not justify factoriza-
121 tion of the joint law, particularly for an ensemble with an unpinned collective
122 phase.

123 A spatial version replaces the finite sum by an integral against a kernel
124 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-
125 field strength is a kernel average, not a quantity that grows with the arbitrary
126 measure of the substrate. Reduction of a heterogeneous spatial kernel to a
127 scalar K is an additional modelling step, and the threshold in Eq. (2) does
128 not automatically apply to every spatial or signed network.

129 **2.2 Physical identification and calibration**

130 Regular mesh refinement gives convergence of a discrete coupling-energy
131 quadrature to its continuum integral: scalar energy convergence, not conver-

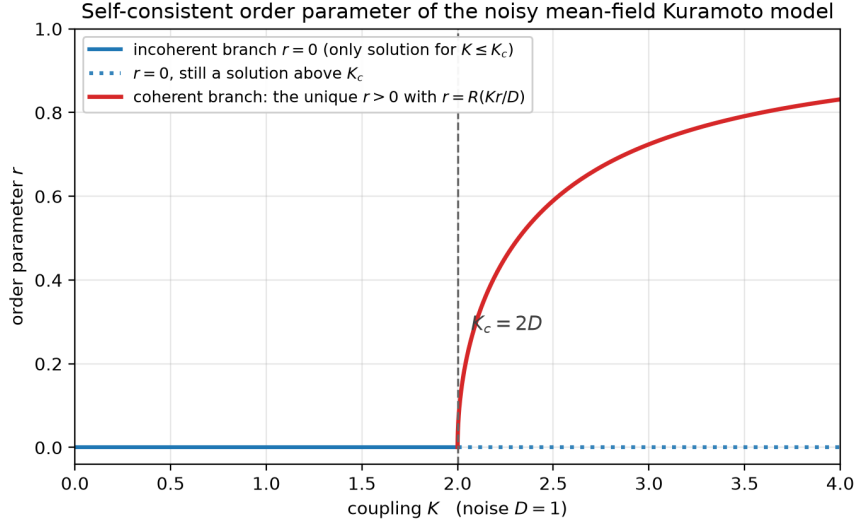


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

132 gence of trajectories and not a construction of a physical kernel. The model
133 posits a continuum kernel and separately identifies its cortical realization
134 with endogenous electromagnetic coupling. The formal predicate requires
135 a jointly continuous kernel, a normalized substrate, a mean-field strength
136 equal to its spatial average, and no single site providing more than half of
137 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
138 model calibrated to cortex.

139 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
140 2010; Anastassiou & Koch, 2015), and field-based models have been proposed

141 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 142 Fadden, 2020). These observations motivate an effective interaction without
 143 establishing its magnitude relative to phase diffusion, its spatial averaging
 144 rule, or its causal role in experience; changes in phase organization under
 145 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 146 coupling threshold.

147 The physical-unit conversion is the sharpest open limitation of the frame-
 148 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
 149 shift and oscillation frequency, is dimensionless. It cannot be compared di-
 150 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

151 requires a rate calibration γ , where s is the time shift per unit field and E
 152 the field amplitude. Supplemental Material sets out why γ is not recoverable
 153 from the other factors: it states how a per-cycle phase advance accumulates
 154 into a drift rate, and the choice that would let it cancel removes the very
 155 ratio the threshold test is about. The saved field-strength calculations are
 156 conditional on this calibration and on the chosen spatial aggregation; they do
 157 not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction
 158 kernel and phase diffusion from the same physical system is necessary before
 159 the threshold becomes a quantitative biological claim; Section 3.2 makes that
 160 measurement the first item of the proposed experiment.

161 3 Recovery after awakening: a conditional 162 prediction

163 Sleep-associated changes in extracellular space and clearance motivate a pos-
164 sible slow control of coupling (Xie et al., 2013). That geometry is measurable
165 and its cortical values are known: the extracellular space occupies a volume
166 fraction $\alpha \approx 0.2$ of tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nichol-
167 son, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al.,
168 2013). It reaches Eq. (4) through the field amplitude E rather than through
169 a further mechanism: below 10 kHz cortical tissue conducts approximately
170 resistively at $0.3\text{--}0.6\text{ S m}^{-1}$, an order of magnitude set by the conductiv-
171 ity of saline diluted into that volume fraction (Barbour, 2017), so at fixed
172 transmembrane current density a larger α yields a smaller extracellular field.
173 We hypothesize that recovery after awakening includes a change in effective
174 coupling $K(t)$, mediated by extracellular geometry, on a slower timescale
175 than phase dynamics. The conductivity route fixes the sign that hypothesis
176 predicts—contraction of the extracellular space on awakening raises K —so
177 the design of Section 3.2 can fail it in direction as well as in magnitude. It
178 fixes nothing further: it is an effective-medium argument at fixed current
179 density, and is silent on how a heterogeneous field aggregates into the scalar
180 K of Eq. (2). Neither the monotonic relation between geometry and cou-
181 pling nor its role in human sleep inertia follows from the cited physiology;
182 the model requires both to be measured.

183 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 184 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

185 The ideal stationary description has a subthreshold interval followed by con-
 186 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 187 monotonic coupling alone does not determine a logistic or any unique sig-
 188 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 189 known. When coupling is inferred from the same coherence trace, this is not
 190 a parameter-free prediction of latency.

191 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 192 threshold, and a finite population carries nonzero fluctuating order below
 193 it. An observable onset therefore requires a predeclared threshold above the
 194 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 195 cusp. Section 4.1 quantifies the limitation.

196 **3.1 Distributional and temporal tests**

197 The stationary model makes a second prediction: phase concentration and
 198 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
 199 cability depends on the stationary-density assumption and the observation
 200 model, and the relation is common to the whole von Mises family, so agree-
 201 ment with it identifies neither ephaptic coupling nor this framework against

202 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
203 $r > 0$, so the limiting threshold ratio follows from the density relation instead
204 of testing it, and ratios near the finite-size floor are especially unstable.

205 The estimators must be independent, which is the constraint most easily
206 violated in practice. Estimating a by inverting the Bessel ratio from r would
207 enforce the relation being tested, and linear statistics on wrapped phases
208 estimate neither quantity.

209 **3.2 A protocol for the awakening experiment**

210 The requirements above define a design rather than a single measurement,
211 and the design is where the content of the hypothesis lives. An adequate test
212 needs all six of the following.

- 213 **1. Coupling and diffusion in the same units.** Estimate K and D
214 from the same cortical regime, in inverse-time units, separating phase
215 diffusion from quenched frequency spread and from observation noise.
216 Without this the threshold $K > 2D$ is not an evaluable statement
217 about cortex.
- 218 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
219 an astrocytic signal is an indicator of the extracellular volume fraction
220 α , not a measurement of K . Its mapping to effective coupling must be
221 established separately, with its own uncertainty, and the conductivity
222 argument of Section 3 supplies only the expected sign of that mapping.

- 223 **3. Phase estimation at adequate spatial resolution.** At least 100
 224 spatial samples per estimate. Estimate r by the circular resultant
 225 length and a by an unweighted regression of log-density on the co-
 226 sine of phase relative to the circular mean. With fewer sites, pooling
 227 over time changes the estimand to a spatiotemporal distribution and
 228 must be stated as such.
- 229 **4. An independent recovery endpoint.** Behavioral recovery measured
 230 separately from the phase trace, so that latency is not defined by the
 231 signal whose latency is at issue.
- 232 **5. A predeclared onset criterion.** A coherence threshold fixed above
 233 the finite-size floor before unblinding, chosen in the knowledge that a
 234 finite ramp smears the cusp.
- 235 **6. Model comparison under matched observation.** Coupling-gated
 236 recovery compared against exponential, flexible sigmoidal and other
 237 slow-recovery models under identical temporal filtering and observation
 238 noise. Latency alone cannot distinguish the mechanism, because many
 239 processes create delays.

240 The design fails informatively in three ways: failure of the stationary
 241 phase-density relation over a sufficiently broad concentration range; absence
 242 of the proposed coupling–geometry association; or a recovery trace better
 243 explained by an alternative after finite-rate and observation effects are ac-
 244 counted for. Two apparent failures are not failures. A linear apparent onset

is evidence against the adiabatic approximation only where that approximation is independently justified, and it does not falsify the field hypothesis under arbitrary ramp speeds. The absence of a fold in the stationary equation excludes stationary coexistence of two positive branches in this model, but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

4 Simulation results

The simulations characterize the finite-size, finite-rate regime that the stationary theory leaves open, and they bear on whether the prediction of Section 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate methodological issues that also arise in analyses of driven transitions and dissipation-driven plasticity. Each is a finite run at its sampled parameters and not a general statement, and none discharges a hypothesis of the composition in Section 5. Each run uses explicit seeds and compact saved summaries, and the numerical values below are generated from those artifacts. The supplement gives parameters, estimators, failure criteria and additional panels. Identical-frequency runs use phase diffusion rather than a quenched frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

4.1 Ramp speed sets the apparent onset exponent

The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at

265 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
 266 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
 267 with a fitted delay exponent 0.443. Their apparent onset exponents run from
 268 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the
 269 exact stationary branch sampled on the same coupling grids. At the fastest
 270 speed the ensemble mean never reaches the declared fit window, so the value
 271 the estimator returns there, 0.964, is a local log-slope of a delayed trajectory
 272 rather than an onset exponent.

273 The exponent recovered by fitting a driven transition is a joint function
 274 of the mechanism and the drive rate. An onset exponent read off a recovery
 275 trace therefore says little about the underlying transition unless the ramp rate
 276 is independently known and the finite-size floor is declared, and that holds
 277 for any such analysis, not only for one conducted under this framework. The
 278 prediction of Section 3 is the first thing it constrains: the stationary square-
 279 root law cannot be imposed on an arbitrary recovery trace, and its absence
 280 is not by itself evidence against a coupling-gated mechanism. The specified
 281 deviation-based critical ramp speed is not bracketed by the sweep.

282 The dynamical-selection control shows escape toward a coherent branch
 283 above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static
 284 value 0.68270. Refining the integration step by a factor of 4 at fixed dura-
 285 tion moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama
 286 bias; the last refinement still moves it by 0.00147, so the remaining -0.00102
 287 bounds the finite-size part rather than measuring it. The fitted growth-rate

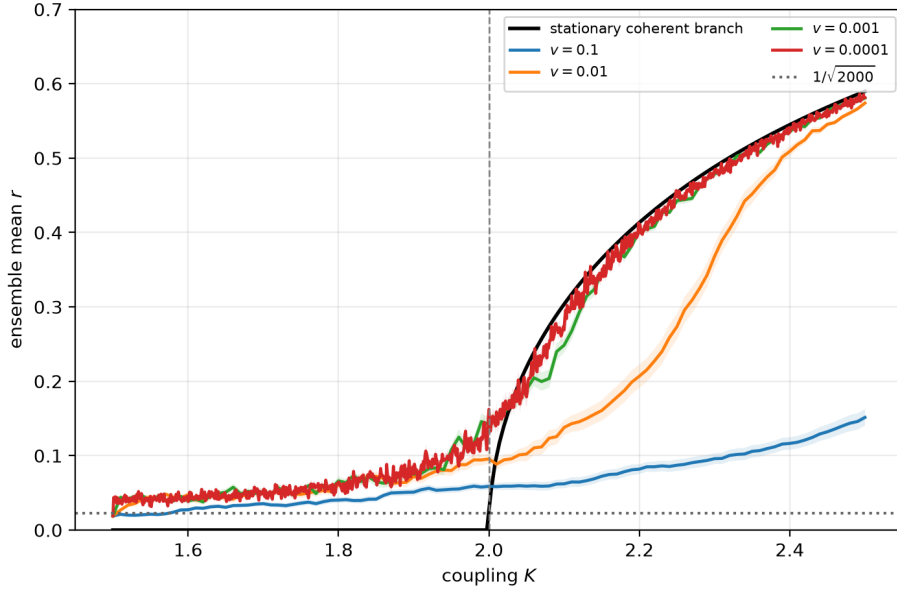


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

288 slope is 0.41505 against the theoretical $1/2$, over a window bounded below
 289 by the finite-size floor and above by the static branch; the same integrator
 290 bias enters that fit. The propagation-of-chaos diagnostics show decreasing
 291 subcritical dependence with increasing population size. In the supercritical
 292 unpinned ensemble, however, pair correlation remains substantial because
 293 realizations share a random collective orientation. This negative uncondi-
 294 tional result does not settle a conditional mean-field limit after removal of
 295 that orientation.

296 4.2 Spatial coupling, frustration and calibration

297 A periodic two-dimensional sheet with row-normalized exponential coupling
298 remains coherent in the sampled empirical decay-length band under the de-
299 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
300 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
301 eter regime; fixing total coupling while changing range tests fragmentation,
302 not whether cortex supplies that total coupling.

303 A row-balanced Dale-law network shows that row balance alone does not
304 guarantee a floor-level incoherent baseline. In a separately specified weak-
305 synapse regime, a uniform modulatory term produces an operational coher-
306 ence crossing inside the bracket $K_{\text{eff}}/D \in [1.958, 2.009]$, from a coupling grid
307 refined to steps of 0.025 inside the one geometric step $[1.857, 2.110]$ that lo-
308 cates it. The refined bracket still contains the model’s own mean-field thresh-
309 old $K_c/D = 2$, so a tenfold finer grid does not separate the crossing from
310 the threshold at this size. Signed coupling amplifies finite-size fluctuations
311 in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which is the
312 direction a resolved crossing would be displaced in; the refinement bounds
313 that displacement by the width of the bracket rather than exhibiting it. The
314 quantity K_{eff}/D is dimensionless; carrying it over to cortex needs the rate
315 calibration γ of Section 2.2, which neither this sweep nor the spatial control
316 supplies.

317 4.3 Squared-drift plasticity does not learn structure

318 The predictive dissipation bound of Section 5 bounds nonpredictive memory
319 by dissipation, and it is tempting to read a system that lowers its dissipation
320 as thereby coming to mirror the structure of its input. The bound licenses no
321 such reading: a memory carrying no information at all satisfies it with room
322 to spare, so lowering dissipation presses toward less representation rather
323 than toward better-aligned representation. This control exhibits a system in
324 which the pressure is not merely toward less but away.

325 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

326 It is a quadratic functional of an affine image of the kernel. The formalization
327 constructs the bounded drift operator A and its adjoint on the relevant L^2
328 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
329 decreases this objective at fixed phases. The objective is not generally a
330 thermodynamic entropy-production rate, which requires probability-current
331 information. Whether biological couplings follow this flow is an additional
332 modelling question.

333 The joint control evolves noisy phases with clustered frequencies and
334 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
335 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
336 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so

337 both endpoints contribute. Resource constraints and evolving phases re-
 338 move the fixed-phase monotonicity guarantee. Each seed runs four arms
 339 on one phase-noise stream: the gradient step, a symmetric random step of
 340 the same Frobenius norm, that same gradient step under a random node
 341 relabelling, and a frozen kernel. Denying an implication requires exhibiting
 342 its antecedent, so the reported quantity is how much of the objective the
 343 update actually removes: symmetric coupling cancels from the mean drift,
 344 which fixes a floor of 990 beneath the frozen arm's 1380.6063 to 1382.3448,
 345 and the gradient arm removes between 0.6773 and 0.6914 of that headroom.
 346 The random arm of matched step size removes none of it, so the descent
 347 belongs to the gradient. Matching the step size does not match the defor-
 348 mation it accumulates: successive isotropic steps cancel, leaving the random
 349 arm at a cumulative norm growth of 1.2879 to 1.3056 against the gradient
 350 arm's 12.4094 to 14.6958, so it barely enters the channel through which clip-
 351 and-rescale concentrates the fixed resource onto fewer edges. The permuted
 352 arm enters that channel. A relabelling is an isometry that leaves the multiset
 353 of step entries untouched, so the arm takes a step of the gradient's size and
 354 of its shape on edges the gradient did not select: it reaches a norm growth
 355 of 11.4185 to 16.3296, bracketing the gradient arm's, and a second-half order
 356 of 0.9184 to 0.9189, which lies inside the gradient arm's range. Its descent
 357 fraction is -0.3319 to -0.1607 : at the gradient arm's own deformation and
 358 coherence, a step that carries the gradient's entries to the wrong edges raises
 359 the objective instead of lowering it. The learning rate 0.2 is the rate at which

the objective descends furthest across a sweep from 0.001 to 0.4 evaluated on one of the three reported seeds, not a rate at which the update is known to converge.

Across three seeds, second-half order lies between 0.9167 and 0.9222: coherence survives the descent. Alignment with the cluster template is read from the ratio of mean within-cluster to mean between-cluster coupling, which is unchanged by rescaling the kernel. That ratio starts between 0.9824 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves it at 0.9965 to 0.9997 and the equally deformed permuted arm at 0.7290 to 1.2935, a range that contains the no-preference line and does not overlap the gradient arm's; against 2000 node relabellings the final kernel is further from the true template than from a fraction 0.9510 to 1.0000 of them (Fig. 3). The reference for that fraction is the frozen arm on the same seeds, which reaches 0.6435 to 0.9110 without any update at all, and not a null median of one half; the increment the gradient step contributes is the difference between those ranges, which at three seeds is small enough that they nearly meet. The permuted arm reaches 0.2680 to 0.8410, so what carries the gradient arm above the frozen reference is the direction of the step and not the deformation it accumulates. Two readouts do not carry the effect and we do not rest it on them. The initial-minus-final Frobenius distance to the template lies between -58.0962 and -48.1760 , so that distance grows; but the kernel norm grows by a factor of 12.4094 to 14.6958 over the same interval, and a fixed total resource constrains the sum of the entries rather than the norm,

383 so most of that growth is concentration of mass onto fewer edges rather than
 384 loss of alignment. The edge-level correlation with the template ends between
 385 -0.0362 and -0.0208 , near its starting value: at this sparsity the edge-level
 386 variance swamps a contrast that the block means resolve. The template is a
 387 covariance of shared latent cluster signals, not an input to the update or a
 388 measured covariance of the constant frequencies.

389 The objective descends by most of what is available to it, coherence is
 390 maintained, and within-cluster coupling falls to a fraction of between-cluster
 391 coupling. Coherence is therefore not a proxy for representational alignment
 392 in this regime, and a plateau in the objective establishes neither stationar-
 393 ity of the adaptive process nor a thermodynamic entropy-production rate.
 394 Nor is the direction of the failure accidental. Symmetric coupling conserves
 395 the mean drift, so minimizing $\sum_i v_i^2$ can only equalize drift across nodes,
 396 and equalizing drift rewards coupling between clusters of unlike frequency—
 397 the complement of a template that rewards coupling within clusters of like
 398 frequency. The objective and the template are in this pairing opposed by con-
 399 struction, and that is the point: nothing about a system’s dissipation fixes
 400 which structure its slow variables acquire. What fails here is the objective
 401 rather than thermodynamically specified learning at large: an Ising model
 402 whose couplings are annealed degrees of freedom, driven by external fields
 403 that carry the data, memorizes and generalizes under ordinary statistical-
 404 mechanical rules (Corberi et al., 2026). Those couplings equilibrate under
 405 a Hamiltonian containing the data; these descend a squared-drift functional

406 whose stationary point the data never enters. Which functional the slow
407 variables follow is what decides whether they acquire environmental struc-
408 ture.

409 **4.4 The estimation protocol demonstrated on public** 410 **EEG**

411 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
412 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
413 et al., 2025). The purpose is to show what the protocol produces and where
414 it is fragile, not to test the awakening hypothesis, for which these data are
415 the wrong design.

416 Two fragilities appear immediately. The implemented estimator pools 62
417 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
418 tion rather than an instantaneous spatial one—a departure from the protocol
419 forced by channel count, and one that any scalp implementation will face.
420 Bipolar and common-reference montages give different residuals, making ref-
421 erencing a material analysis choice rather than a preprocessing detail.

422 The bipolar results are compatible with the Bessel relation over the ob-
423 served concentration range. That range is too narrow to distinguish the
424 curve from its linear approximation at the observed scatter, so the check is
425 uninformative about the relation itself. An informative test needs a broader
426 concentration range and explicit treatment of source mixing, finite samples

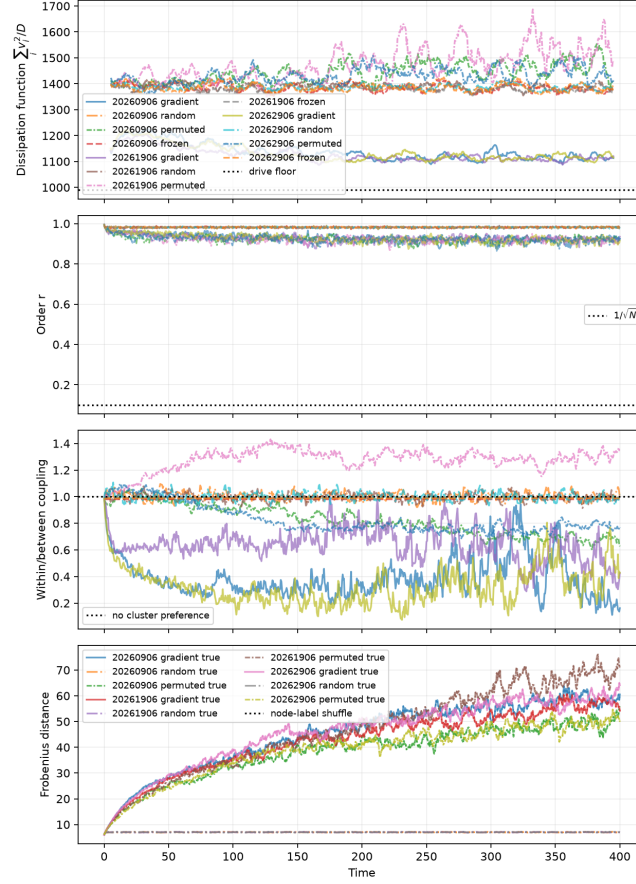


Figure 3: Joint fast/slow dynamics, four arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), the gradient step under a random node relabelling (dash-dot-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation against the drive floor the conserved mean drift imposes; only the gradient arm descends. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line, the relabelled arm stays above the gradient arm at the same kernel deformation, and the remaining arms stay on the line. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

427 and temporal pooling; intracranial or high-density recordings across a gen-
428 uine state transition are the natural setting.

429 5 Thermodynamic constraints and the for- 430 mal composition

431 The synchronization model of Section 2 is one link of a longer conditional
432 construction, whose nine nodes are:

433 **n1** a finite phase space has bounded information capacity; **n2** a
434 broken symmetry forces a boundary, where the field leaves its vac-
435 uum manifold; **n3** the boundary erases, and erasure costs heat;
436 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
437 couplings coarse-grain to a continuum limit; **n6** a specified phys-
438 ical field realizes that limiting coupling; **n7** above threshold the
439 field carries a unique coherent order parameter; **n8** compatible
440 local states glue to a unique global section; **n9** that section is the
441 unique fixed point of a self-prediction map.

442 Node-level results are established or cited; what the composition consumes
443 are the eight implications between them. Figure 4 shows where they sit and
444 what kind of assumption each one is, and Table 1 states them, classified by
445 their role in the composition rather than by their plausibility. In particular,
446 finite capacity does not imply a symmetry-breaking boundary: **n1** and **n2** are
447 independent roots, even though the composition can accept an implication
448 between them as an argument.

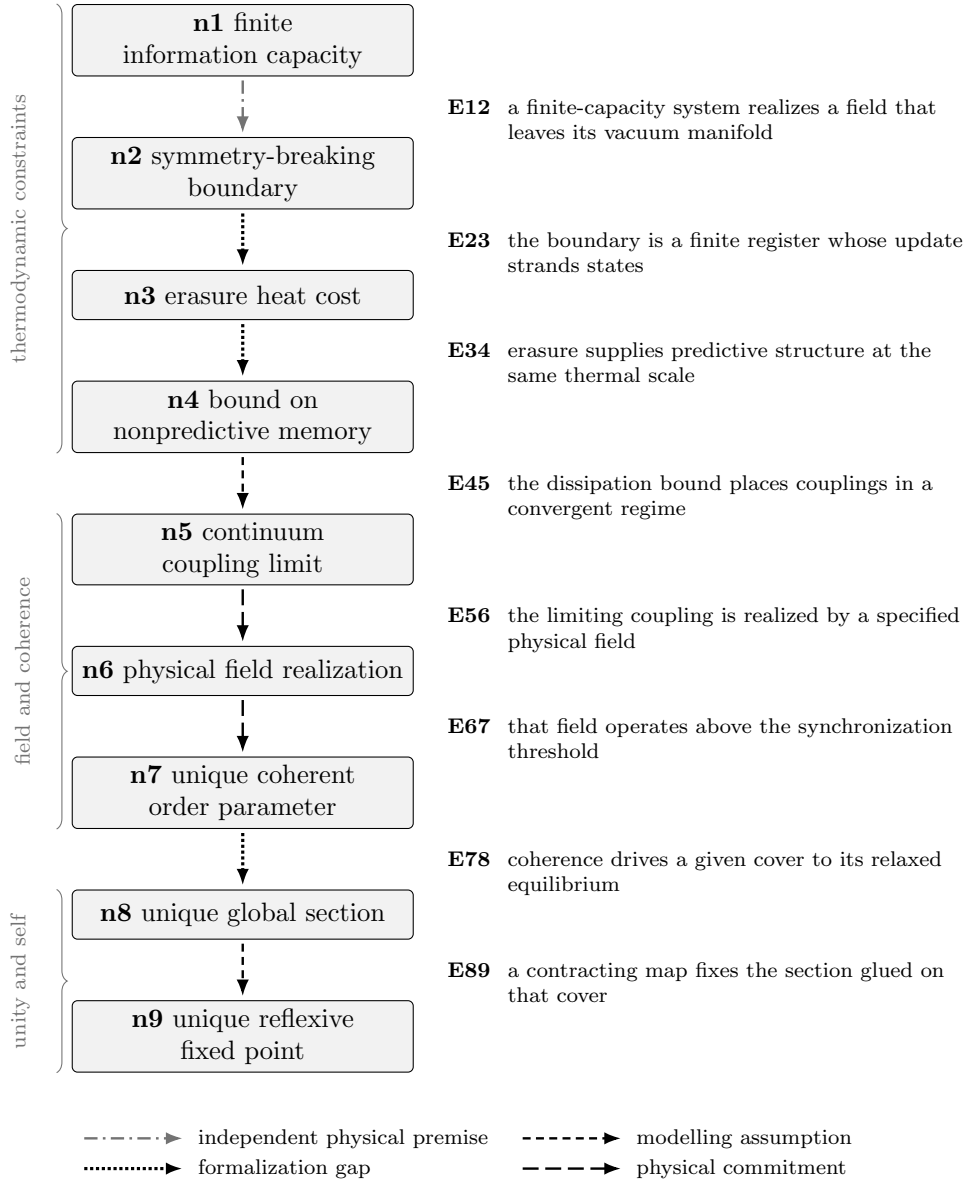


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12	Independent	A finite-capacity system realizes a field that
capacity $\rightarrow$	physical	leaves its vacuum manifold. Capacity alone
boundary	premise	does not imply this.
E23	Formalization	The boundary is associated with a finite
boundary $\rightarrow$	gap	register whose update leaves states
erasure		unreachable.
E34	Formalization	Erasures supplies a predictive structure with
erasure $\rightarrow$	gap	the same thermal scale and heat budget,
prediction		and nonzero wasted memory.
E45	Modelling	A predictive dissipation bound places
prediction $\rightarrow$	assumption	coupling energies in a convergent
continuum		coarse-graining regime.
E56	Physical	The limiting coupling is realized by a
continuum $\rightarrow$	commitment	specified field satisfying the kernel
field		conditions. Its identification with cortex is
		empirical.
E67	Physical	That field operates above the
field $\rightarrow$	commitment	synchronization threshold.
coherence		
E78	Formalization	Coherence drives a cover, supplied as a
coherence $\rightarrow$	gap	separate argument of the theorem, to an
unity		equilibrium reached by relaxation at
		coupling at least the named mean-field
		value. That its patches carry compatible
		local states is a field of the cover rather
		than a conjunct of this edge.
E89	Modelling	The state glued on that same cover is fixed
unity $\rightarrow$ self	assumption	by a contracting map mediated by its local
		self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

449 5.1 Capacity, boundaries and predictive memory

450 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
 451 equality only for the uniform distribution. A stream with more possible
 452 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 453 a continuous field on a connected substrate that reaches distinct components
 454 of a disconnected vacuum manifold must leave that manifold somewhere,
 455 the simplest topological-defect construction (Kibble, 1976). Relating this
 456 locus to a particular finite information-processing register is the additional
 457 hypothesis E12 carries.

458 A non-surjective update on a finite register is non-injective. Under the
 459 specified system–bath model and thermal assumptions, erasure carries a Lan-
 460 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 461 update erases, and logical many-to-one processing must be distinguished from
 462 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 463 tion records the physical heat identity as an obligation of each model and
 464 supplies an explicit one-bit eraser satisfying it.

465 For a system state X_t driven by an environmental signal S_t , the predictive
 466 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

467 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 468 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains

nonpredictive memory at a given dissipation budget. It does not specify a learning rule, maximize prediction, or require the internal representation to reproduce environmental geometry—the point the plasticity control of Section 4 makes concretely. Connecting this budget to Landauer heat additionally requires nonpredictive information to be bounded by the entropy erased by the register. The formal bridge derives Eq. (7) from Landauer’s bound under that identification; it does not derive the identification.

A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand side tends to zero, and Gibbs’ inequality proves nonnegativity of relative entropy, not a general identification of relative entropy with physical entropy production. The chain uses a different technical sense of information at each step—a counting bound on a register, a thermodynamic heat cost, a predictive mutual information—and those senses are not interchangeable (Rosas et al., 2026). These distinctions prevent an informational inequality from being read as a derivation of environmental learning, Bayesian inference, or semantic content.

5.2 Compatible states and reflexive fixed points

A sheaf formalizes local descriptions and their restriction to smaller regions. If local sections agree on every overlap, the sheaf condition gives a unique global section. In the probability-sheaf construction, this is consistency of local density information, and it is not implied by agreement of oscillator phases: the development includes a counterexample with equal phases but

491 incompatible densities on an overlap. Section 6 rests on this distinction,
492 and the model excludes such configurations only through its compatibility
493 hypothesis.

494 A thermodynamic cover couples this gluing construction to a phase con-
495 figuration at a potential minimum. For the deterministic positive-coupling
496 model, convergence to that minimum is proved under an explicit initial basin
497 condition; other equilibria prevent an unconditional conclusion. The station-
498 ary noisy order parameter, the deterministic relaxation, and the cover must
499 therefore be connected by E78 rather than identified without a hypothesis.
500 Neither field continuity nor synchronization alone guarantees a globally com-
501 patible representation.

502 A reflexive extension equips global sections with a complete metric and a
503 predictive map formed by restricting a state to a local encoding region and
504 reading that encoding out. If the map contracts and fixes the section supplied
505 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
506 The contraction rate is assumed, and the equality between the glued section
507 and the fixed point is part of E89. A coherent order parameter supplies the
508 supercritical inequality required by the stated contraction bound; it does not
509 construct the map or prove that cortical dynamics implements it.

510 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
511 from two structures it also takes as arguments, a thermodynamic cover and
512 a reflexive boundary. The eight edges are not the whole of what is assumed:
513 the cover’s overlap-compatibility field—the load-bearing assumption of the

514 unity-is-not-synchrony argument—and the boundary’s resonance condition
 515 are class fields, so they are consumed by the theorem without appearing
 516 among its edges. A single witness satisfies all eight at once, so their conjunc-
 517 tion is not empty; it is assembled from a double well, a register, a uniform-grid
 518 mesh and a three-site cortex, which remain distinct toy systems rather than
 519 one mechanism. The project declares no additional Lean axioms beyond
 520 the foundational axioms used by Lean’s libraries; physical content resides
 521 in hypotheses and model fields. This checks the conditional mathematical
 522 composition, not the empirical truth of its premises. Detailed constructions,
 523 scope statements and counterexamples are retained in the supplement and
 524 source files.

525 **6 What the construction commits to about** 526 **consciousness**

527 The composition terminates in a mathematical object, and we take a posi-
 528 tion on what that object is. The commitment has two clauses. *Unity*: the
 529 unique global section glued from compatible local states is the formal corre-
 530 late of the unity of a conscious episode—the fact that a subject undergoes
 531 one experience rather than a collection of co-occurring local ones. *Selfhood*:
 532 the fixed point of the self-prediction map on that same section is the formal
 533 correlate of a minimal self, a state that is what it takes itself to be. Neither
 534 clause is a theorem, and no simulation here supports either. We argue for

535 them, then say what would count against them.

536 Three features of the construction motivate the identification, and each
537 distinguishes it from a bare synchrony measure.

538 First, compatibility rather than agreement. Oscillator phases can agree
539 everywhere while the local probabilistic descriptions they carry remain in-
540 compatible on their overlaps, and Section 5.2 exhibits that configuration
541 inside the formalization, not only as an illustration. If the unity of experi-
542 ence is a matter of local contents being about one world rather than merely
543 co-timed, then compatibility, not coherence, is the property to look for—and
544 coherence enters as the physical condition under which compatibility can be
545 maintained, which is what E78 asserts.

546 Second, uniqueness follows from the construction rather than from a sep-
547 arate selection principle. Accounts of conscious unity need something that
548 picks one experience from among the candidates a system could support;
549 integrated information theory postulates exclusion for this purpose (Tononi
550 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
551 whose local sections agree, the sheaf condition gives exactly one global sec-
552 tion, and given a contracting self-prediction map, Banach’s theorem gives
553 exactly one fixed point. The scope of that claim needs stating exactly, be-
554 cause the stronger reading is available and is not ours. Uniqueness is relative
555 to a cover and a map, and choosing them—which regions, which overlaps,
556 which encoding region, and so where the subject ends—is not supplied by ei-
557 ther theorem. It is what E78 and E89 carry. What the construction removes

558 is the need for an additional principle to select among the experiences a fixed
559 cover and map would otherwise admit; it does not remove the need to say
560 which cover and map a given subject has. That is a structural argument,
561 not evidence.

562 Third, the reflexive clause makes the self a condition rather than an addi-
563 tion. The predictive map restricts a global state to a region encoding it and
564 reads that encoding back out, so a fixed point is a state whose self-model is
565 accurate about the state it models. That condition is available only because
566 the state modelled is a single glued object: a system whose local descrip-
567 tions do not cohere has no such state for a self-model to be about. Unity
568 and selfhood are therefore one construction read at two levels, which is why
569 `UnifiedSelf` requires the fixed point to be the glued section itself rather
570 than an unrelated section.

571 The costs belong with the argument. The commitment does not address
572 why any physical condition is accompanied by experience; it is a proposal
573 about which physical state is the conscious one, not about why states of that
574 kind feel like anything. It is therefore open to the objection that a system
575 could satisfy every condition—compatible local states, a supercritical field,
576 an accurate self-model—while nothing is experienced. We have no reply to
577 that objection and claim none. The identification is also multiply realizable
578 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
579 or to electromagnetic coupling, so the substrate claim is carried entirely by
580 E56 and E67 and is empirical. No conclusion about artificial systems follows

581 from the formal results.

582 What could count against the commitment is dissociation. It predicts that
583 the two clauses and the presence of a unified conscious episode go together,
584 so a state in which local cortical descriptions are demonstrably compatible
585 and the field supercritical, while the subject reports no unified experience,
586 would tell against it; so would unified reported experience where compati-
587 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
588 states are the places to look, because they separate coherence from report
589 in ways ordinary waking does not. Such tests need an operational measure
590 of overlap compatibility on neural data, which this article does not supply;
591 constructing one is the main obstacle to making the commitment empirically
592 live rather than merely stated. Supplemental Material specifies what such an
593 observable would have to satisfy—a restriction map, a statistic with a null
594 under decoding noise, validation against configurations whose compatibility
595 is built rather than inferred—and the three ways it would fail silently, one
596 of which is reading compatibility off synchrony and so assuming away the
597 distinction being tested.

598 The physical tests of Sections 3 and 4 are independent of all of this. They
599 concern coupling, coherence and recovery, and can fail while the identification
600 stands, or succeed while it is wrong.

601 7 Discussion

602 The framework connects thermodynamic information constraints, collective
603 phase organization and compatible local representations, conditionally and
604 with its assumptions named. Its biological content is a coupling-gated re-
605 covery hypothesis with defined observables, a stated protocol, and failure
606 conditions declared in advance. Its formal content is a composition whose
607 eight gaps are exhibited rather than argued past, and which can be inhabited
608 by a single model. Its interpretive content is the identification of Section 6,
609 offered as a commitment with reasons.

610 The simulations are useful because they restrict the hypothesis rather
611 than support it. Finite ramps smear the stationary onset to the point where
612 a fitted onset exponent chiefly reports the ramp; an unpinned collective
613 phase makes independence diagnostics ambiguous in exactly the supercritical
614 regime the model cares about; and squared-drift plasticity maintains coher-
615 ence while degrading structural alignment. The second and third of these
616 are results about method and about a common intuition, and they transfer
617 beyond this framework.

618 The proposal differs in emphasis from accounts based on integrated in-
619 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
620 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
621 candidate interaction medium and a mathematical compatibility condition,
622 and, relative to a given cover and self-prediction map, it derives the unique-

ness those accounts postulate. It does not derive the exclusion principle of integrated information, identify every broadcast with a field transition, or derive a generative model from a dissipation bound. Continuous fields can carry incompatible local descriptions and discrete models can admit sheaf constructions, so continuity alone does not settle the combination problem; compatibility is what this account offers in its place.

Two formal results bear on the substrate question, and both admit a stronger reading than they support. At matched coupling resource, a fixed wiring support that excludes a better-correlated pair can be suboptimal—a statement about constrained optimization, not about biology. On an atomless substrate, a kernel supported on finitely many points contributes zero to the chosen continuum coupling integral. That is a property of the measure and the observable, and it cuts against this framework’s own coarse-graining step as much as against anything else. It does not imply that digital hardware cannot approximate a field or support consciousness, and no biological-versus-artificial conclusion follows.

The empirical priorities follow directly from Section 3.2: estimate effective coupling and diffusion in consistent units, test the proposed relation between coupling and extracellular geometry, and measure recovery against an independently defined behavioral endpoint under matched model comparison. A favorable concentration–coherence relation would support a statistical description, not identify its physical cause; an unfavorable result in a regime satisfying the stated assumptions would constrain that description.

646 Successful synchronization simulations cannot substitute for measurements
647 distinguishing ephaptic interactions from other coupling mechanisms.

648 The article therefore asks to be judged in three separable parts: a pre-
649 diction that can be tested, a composition that can be checked, and an iden-
650 tification that can be argued about. Keeping them separate is what makes
651 the whole assessable.

652 **Data and code availability**

653 The Lean development, simulation scripts, tests and saved numerical sum-
654 maries are maintained in the manuscript source repository. Publication sim-
655 ulation values are regenerated from saved summaries without rerunning pro-
656 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
657 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
658 in Supplemental Material. The source repository contains the reproducible
659 analysis and does not redistribute the raw recordings.

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661 This research received no external funding. The author declares no compet-
662 ing interests.

663 Use of AI tools

664 AI assistance to this work was substantial and is not separable by task.
665 Several coding assistants and chat interfaces were used in an interleaved way
666 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
667 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
668 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
669 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
670 they contributed to the Lean development, the simulation and analysis code,
671 and the drafting and revision of both documents. Because the tools were
672 used in combination and in alternation, no section, proof, figure or result is
673 attributable to any one of them, and none is claimed to be. The author is
674 responsible for all content, including everything a tool produced. Machine
675 checking establishes the stated Lean results under their hypotheses; it does
676 not validate biological interpretations or replace scientific review of generated
677 code and prose.

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